CHAPTER

16

Windowed-Sinc Filters

Windowed-sinc filters are used to separate one band of frequencies from another. They are very stable, produce few surprises, and can be pushed to incredible performance levels. These exceptional frequency domain characteristics are obtained at the expense of poor performance in the time domain, including excessive ripple and overshoot in the step response. When carried out by standard convolution, windowed-sinc filters are easy to program, but slow to execute. Chapter 18 shows how the FFT can be used to dramatically improve the computational speed of these filters.

Strategy of the Windowed-Sinc

Figure 16-1 illustrates the idea behind the windowed-sinc filter. In (a), the frequency response of the *ideal* low-pass filter is shown. All frequencies below the cutoff frequency, f_C, are passed with unity amplitude, while all higher frequencies are blocked. The passband is perfectly flat, the attenuation in the stopband is infinite, and the transition between the two is infinitesimally small.

Taking the Inverse Fourier Transform of this ideal frequency response produces the ideal filter kernel (impulse response) shown in (b). As previously discussed (see Chapter 11, Eq. 11-4), this curve is of the general form: $\sin(x)/x$, called the **sinc function**, given by:

$$h[i] = \frac{\sin(2\pi f_C i)}{i\pi}$$

Convolving an input signal with this filter kernel provides a *perfect* low-pass filter. The problem is, the sinc function continues to both negative and positive infinity without dropping to zero amplitude. While this infinite length is not a problem for *mathematics*, it is a show stopper for *computers*.